



Introduction to NLP

AIAA 4051 Lecture 3

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Part-of-Speech of English

- In grammar, a part-of-speech (POS) is a category of words
 - Also known as word class[1] or grammatical category[2]
 - Words with the same POS have similar grammatical properties.
- List of major POS tags
 - Noun (NN)
 - Pronoun (NNP)
 - Verb (VB)
 - Adjective (JJ)
 - Adverb (RB)
 - Article or Determiner (DT)
 - Preposition (IN)
 - Conjunction (CC)

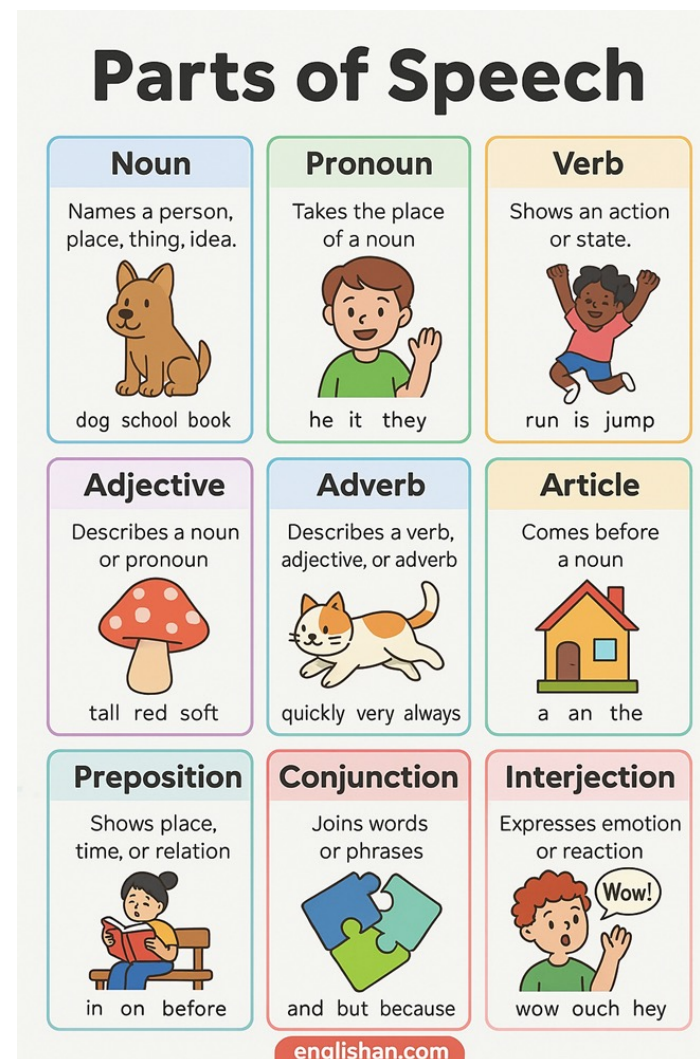


Image source: <https://englishan.com/wp-content/smush-webp/2019/10/Parts-of-Speech-with-Examples.png.webp>

A running example

- A line from one of my favorite movies "The Hateful Eight"



"He (**PRP**) slowly (**RB**) looked (**VBD**) over (**RP**) the (**DT**) cold (**JJ**) coffee (**NN**) in (**IN**) Wyoming (**NNP**), and (**CC**) she (**PRP**) might (**MD**) be (**VB**) a (**DT**) lying (**JJ**) prisoner (**NN**)."

- The meat: nouns, verbs, adjectives
 - **Wyoming**: NNP a special place.
 - **coffee**: NN, a mass (uncountable) noun
 - **looked**: Verb in pass tense (VBD)
 - **cold**: Adjective
- The Glue: adverb, proposition, determiner
 - **slowly**: Adverb modifying the verb looked.
 - **over**: Particle that change the meaning of "looked".
 - **the**: pointing to specific coffee (rather than any coffee).

Image source: https://thehatefuleight.fandom.com/wiki/Marquis_Warren?file=Download_%2810%29.jpg

Why POS tags are important

- Syntactic information: how words are ordered in a sentence.

- noun-verb: she laughs
- determiner-noun: a rabbit
- adjective-noun: high building
- verb-adverb: move faster
- preposition-noun: on tables

He rode his **hoarse** (NN) across the **desert** (NN).

$$P(JJ|PRP\$) \times P(IN|JJ) \approx 0$$

He rode his **horse** (NN) across the **desert** (NN).

$P(NN|PRP\$) \times P(IN|NN)$ is more significant

Misspelled Word	POS Tag	Intended Word	Correct POS	Example
There	RB/EX	Their	PRP\$	"It is there (RB) gun" → their (PRP\$)
Passed	VBD	Past	NN/JJ/IN	"In the passed (VBD)..." → Past (NN)
Effect	VB	Affect	NN	"The cold effect (NN) him" → Affect (VB)
Loose	JJ	Lose	VB	"Don't loose (JJ) the key" → Lose (VB)

Why POS tags are important

- Semantic information: meaning of a word in a context. Useful for:
 - machine translation (building a building): building (verb vs. noun)-> (建 vs. 楼)
 - Recognizing the correct POS help determine the semantics of building.
 - relation extraction (Bill Gates founded MS): gates (verb vs. noun)
 - If Bill Gates is Proper Noun, then there is relationship between Bill Gates and MS
 - If Gates is a verb, then no relationship between Bill Gates and MS.
 - event extraction (They went to a concert): concert (verb vs. noun)
 - If concert is recognized as an noun, then there is an event.
 - entity extraction (I will visit DC): DC?
 - Recognizing DC as a Proper noun.

Why POS tagging is difficult

- Lexical ambiguity

Word	Possible Tags	Examples
"Back"	RB, NN, JJ, VB	"Step back (RB)" vs. "His back (NN) hurts" vs. "The back (JJ) door."
"Like"	IN, VB, JJ	"I like (VB) coffee" vs. "He acts like (IN) a killer."
"Fast"	JJ, RB, NN	"A fast (JJ) horse" vs. "He ran fast (RB)."

- Most models can achieve 97% accuracy [1]

- Most tags (~85%) are not ambiguous

- Janet is always NNP, hesitantly is always RB

- Ambiguous words (~15% of the vocabulary) are common

- 55-67% of word tokens in running text are ambiguous.

- "Back", "Like", "Fast" ...

- Most-frequent-tag baseline: 92%
- Human baseline: 97%

Types:		WSJ	Brown
Unambiguous	(1 tag)	44,432 (86%)	45,799 (85%)
Ambiguous	(2+ tags)	7,025 (14%)	8,050 (15%)
Tokens:			
Unambiguous	(1 tag)	577,421 (45%)	384,349 (33%)
Ambiguous	(2+ tags)	711,780 (55%)	786,646 (67%)

Figure 17.4 Tag ambiguity in the Brown and WSJ corpora (Treebank-3 45-tag tagset).

[1] SLP3 (2026 version), Chap 17

Image source: Figure 17.4 from SLP3 (2026 version).

Fun facts

• The "Buffalo" Problem^[1]

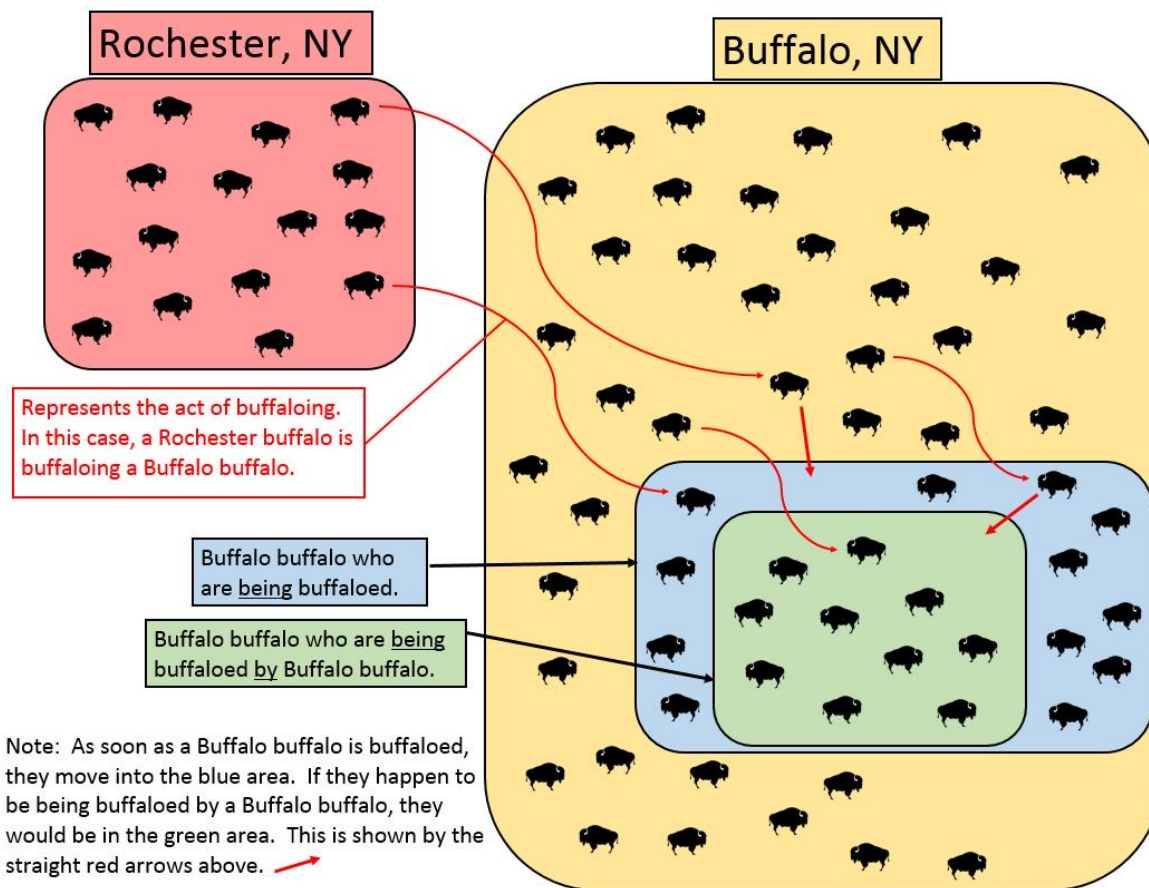
Because of POS ambiguity, the sentence "*Buffalo buffalo Buffalo buffalo buffalo Buffalo buffalo*" is a grammatically correct sentence in English.

[1]https://en.wikipedia.org/wiki/Buffer_buffalo_Buffalo_buffalo_buffalo_Buffalo_buffalo

Image source:

https://upload.wikimedia.org/wikipedia/commons/a/a1/Buffer_buffalo_Buffalo_buffalo_buffalo_buffalo_Buffalo_buffalo_sentence_diagram.jpg

This sentence is laid out below, with the subject of the sentence (Buffalo buffalo Buffalo buffalo buffalo) being completely contained within the green area. They are buffaloeing any of the buffalo contained within the yellow, blue, or green areas (object of the sentence). However, the moment any of those buffalo are buffaloeed, they would also reside in the green area. Therefore, pictorially, the green area contains the subject and object of the sentence.



Fun facts

- **Social Media Chaos:** In modern NLP (Twitter/X), we had to invent new tags!
 - #NLP (Hashtag)
 - @Professor (At-mention)
 - RT (Retweet)
 - lol (Interjection/Emoji)
- POS is just a intermediate task in NLP and is now less needed given LLM.
 - But it is still useful to build intuition about NLP.

Trending in USA

#MyFavoriteRadioMemories

Trending in USA

#TheProudFamily

22.5K Tweets

Trending in USA

#HeyGodCouldYou

2,183 Tweets

Trending in USA

#TongueOutTuesday

2,427 Tweets

The new tags are called “out-of-vocabulary” (OOV) words.

Can be partly addressed by language models that use sub-word.

Hidden Markov Models: motivation

- The ambiguity of POS can be mitigated using more context information.
 - His (PRP\$) **back** (NN) hurts (VBZ)
 - What information helps to decide NN tag for “**back**”?
 - “back” can potentially be only NN, RB, JJ, VB tags, and cannot be DT, CC, NNP, etc.
 - $P(\text{“back”}|\text{NN}) > P(\text{“back”}|\text{DT})$, according to estimated “**emission**” probabilities.
 - further looking at the previous tag PRP\$ narrows it down to NN
 - $P(\text{NN}|\text{PRP\$}) > P(\text{DT}|\text{PRP\$})$: in English, it common to say “someone’s NN” than “someone’s the”
 - This is called the “**transition**” probabilities.
 - The unambiguous tag of “His” makes POS tagging easier.

* PRP\$: Possessive Pronoun

** VBZ: Verb, 3rd Person Singular Present Verb, 3rd Person Singular Present

Hidden Markov Models: formulation

- Want to find the best sequence of POS tags for a given sentence.
- Vocabulary V
- Set of N POS tags $S = \{s_1, \dots, s_n\}$

- Observed sentence $O = [o_1, \dots, o_T], o_t \in V$

All we gotta do is go around the corner

- Hidden states: $Q = [q_1, \dots, q_T], q_t \in S$

q_1 q_2 q_9

- MAP (maximum a posterior) prediction

$$Q^* = \arg \max_Q \Pr(Q|O)$$

DT PRP VBN VB VBZ VB IN DT NN

- Difficulty: there are exponentially many possible sequences.

Hidden Markov Models: formulation

- Bayes rule: $\Pr(Q|O) = \frac{\Pr(O|Q)\Pr(Q)}{\Pr(O)}$
 - Predicting the optimal tags become $Q^* = \arg \max_Q \Pr(Q|O) = \arg \max_Q \Pr(O|Q)P(Q)$
 - The observed input is fixed w.r.t. Q .
 - It is common to drop the normalizing terms in ML/AI optimization.
 - Cannot maximize only one of the two terms:
 - If only maximizing $\Pr(O|Q)$: pick the tags best suits the observation, but not considering how likely the tags are without seeing the words.
 - His (PRP\$) **back** (RB) hurts (VBZ): RB is the most likely tag for back, but won't follow PRP\$.
 - If only maximizing $\Pr(Q)$: pick the best tag without considering the observed words
 - DT JJ NN: this may generating common sentences like "the red car", but are irrelevant.

| n-gram language models: theory

- Two probabilities

- Generating the tag sequence according to following joint distribution

$$\Pr(Q) = \Pr(q_1, \dots, q_T) = \Pr(q_1)\Pr(q_2|q_1)\Pr(q_3|q_1, q_2) \dots \Pr(q_T|q_1, q_2, \dots, q_{T-1})$$

- Emitting the word sequence according to following conditional distribution

$$\Pr(O|Q) = \Pr(o_1, \dots, o_T|q_1, \dots, q_T) = \Pr(o_1|Q)\Pr(o_2|o_1, Q)\Pr(o_3|o_1, o_2, Q) \dots \Pr(o_T|o_1, \dots, o_{T-1}, Q)$$

- Both probabilities requires exponentially many parameters

- $\Pr(q_2|q_1)$: needs $|S|^2$ parameters, one for each combination of (q_1, q_2) pairs.
- $\Pr(q_3|q_1, q_2)$: needs $|S|^3$ parameters, one for each combination of (q_1, q_2, q_3) triples

| The Markov assumption

- What happens **next** depends on the **current** status, regardless of how you arrived at the current status from the **past**.



- Some life examples:
 - To drive on the street, only check the traffic light now (Green, Red, Orange), but don't worry what it was before this moment.
 - To predict if it will rain in the next minute, only need to check the cloud on the sky now, but don't worry how the cloud gathered before.
- The first NLP researcher: Andry Markov (Russian Mathematician 1913)
 - Invented Markov model for writing poems.
 - Manually counted 20,000 letters from Aleksandr Pushkin's novel Eugene Onegin to prove that the probability of a vowel occurring depends on whether the preceding letter was a vowel or a consonant.

The Markov assumption

- For modeling $\Pr(Q)$, we make the Markov assumption:
 - Only care about the previous tag (forget the history) when generating the current

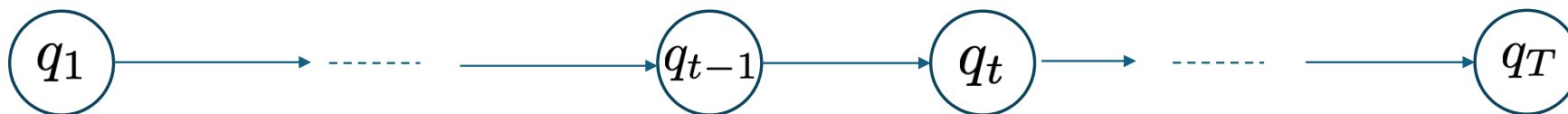
$$\Pr(q_t|q_1, \dots, q_{t-1}) = \Pr(q_t|q_{t-1})$$

Complexity: $|S|^2$

- Simplify the joint distribution

$$\begin{aligned}\Pr(Q) &= \Pr(q_1, \dots, q_T) = \Pr(q_1)\Pr(q_2|q_1)\Pr(q_3|q_1, q_2) \dots \Pr(q_T|q_1, q_2, \dots, q_{T-1}) \\ &= \Pr(q_1)\Pr(q_2|q_1)\Pr(q_3|q_2) \dots \Pr(q_T|q_{T-1})\end{aligned}$$

- The chain of conditionals can be represented using a graphical model
 - A node represent a random variable.
 - An edge represents a dependency, and no edge means conditional independence.



The Markov assumption

- Transition probability matrix A

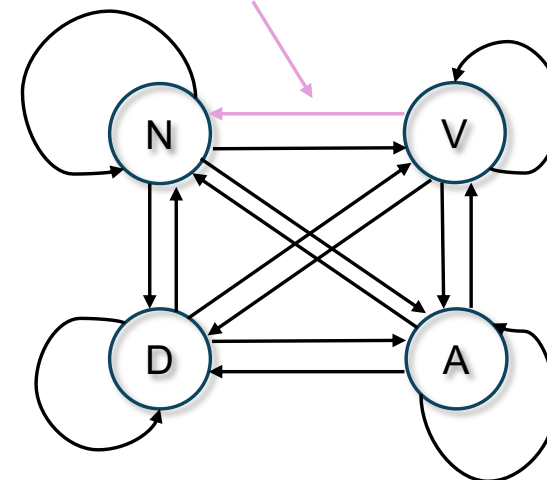
A_{ij} = Probability of going from POS tag i to POS tag j

Current ↓ \ Next →	D (Determiner)	A (Adjective)	N (Noun)	V (Verb)
D (Determiner)	0.05	0.45	0.45	0.05
A (Adjective)	0.05	0.15	0.75	0.05
N (Noun)	0.1	0.05	0.2	0.65
V (Verb)	0.3	0.1	0.5	0.1

Each row sums to 1

$$\Pr(N|V) = 0.5$$

N = Noun
V = Verb
D = Determiner
A = Adjective



- Why it makes sense?

$D \rightarrow A/N$ (0.45+0.45)	"the big ", "a dog "
$A \rightarrow N$ (0.75)	"red car ", "happy child "
$N \rightarrow V$ (0.65)	"dog runs ", "she eats "

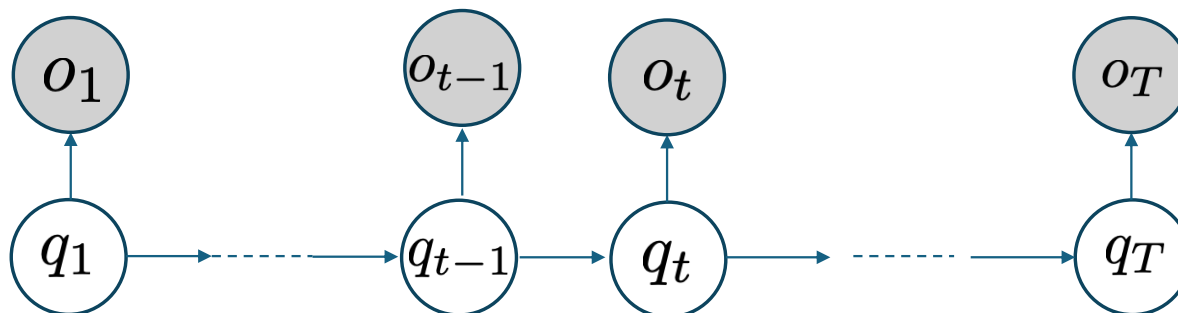
$V \rightarrow N$ (0.50)	"eat apple ", "see bird "
$V \rightarrow D$ (0.30)	"see the... ", "like a... "
$N \rightarrow N$ (0.20)	"computer science ", "coffee cup "

The Markov assumption

- Emission probability
 - A word is emitted based on the current POS tag.

$$\begin{aligned}\Pr(O|Q) &= \Pr(o_1, \dots, o_T | q_1, \dots, q_T) = \Pr(o_1 | Q) \Pr(o_2 | o_1, Q) \Pr(o_3 | o_1, o_2, Q) \dots \Pr(o_T | o_1, \dots, o_{T-1}, Q) \\ &= \Pr(o_1 | q_1) \Pr(o_2 | q_2) \Pr(o_3 | q_3) \dots \Pr(o_T | q_T)\end{aligned}$$

- Remove the following probabilistic dependencies
 - the current word and previous words;
 - the current word and past POS tags;
 - the current word and future POS tags;



The Markov assumption

- Emission probability matrix B

B_{io} = Probability of emitting word o from POS tag i

Tag ↓ \ Word →	the	a	big	red	dog	cat	car	run	eat	see	go
D (Determiner)	0.6	0.3	0.01	0.01	0.02	0.02	0.02	0.005	0.005	0.005	0.005
A (Adjective)	0.01	0.01	0.35	0.3	0.03	0.03	0.04	0.03	0.03	0.03	0.04
N (Noun)	0.02	0.03	0.05	0.04	0.25	0.2	0.18	0.04	0.04	0.06	0.04
V (Verb)	0.005	0.005	0.02	0.02	0.05	0.05	0.05	0.25	0.2	0.18	0.18

- Why it makes sense?

D	<i>the</i> (0.60), <i>a</i> (0.30)	The frequency of <i>the</i> is much higher than <i>a</i>
A	<i>big</i> (0.35), <i>red</i> (0.30)	Most frequent adjectives
N	<i>dog</i> (0.25), <i>cat</i> (0.20), <i>car</i> (0.18)	Most frequent nouns
V	<i>run</i> (0.25), <i>eat</i> (0.20), <i>see/go</i> (~0.18)	Most frequent verbs

The Hidden Markov Model (HMM)

- Prior

$$\Pr(Q) = \Pr(q_1)\Pr(q_2|q_1) \dots \Pr(q_T|q_{T-1}) = \Pr(q_1) \prod_{t=2}^T \Pr(q_t|q_{t-1})$$

with the starting probability $\Pr(q_1) = \pi_{q_1}$ = probability of using q_1 as the first POS tag.

- Likelihood

$$\Pr(O|Q) = \prod_{t=1}^T \Pr(o_t|q_t)$$

- Posterior

$$\Pr(Q|O) = \frac{\Pr(O|Q)\Pr(Q)}{\Pr(O)}$$

Three tasks regarding HMM:

1. **Estimation**: how to estimate the starting, transition, and emission probabilities?
2. **Inference**: How to estimate the probability of a sentence?
3. **Prediction**: How to predict the POS tags of words in a sentence?

Next two lectures will address these tasks.

| Extra readings

- For POS tagging, see Chap 17.1-17.3 of SLP3.
- For HMM modeling, see Chap 17.4 of SLP3.

| Demon

- First generate a corpus of limited length using LLM.
- Specify the set of POS tags.
- Estimate the parameters of an HMM models from the corpus, using relative frequencies only.
- Visualize transition matrix.

Conclusion

- Next: forward and backward algorithms (**dynamic programming**) for HMM.
 - It would be great if you can review basic concepts of dynamic programming.
 - Optimal substructures.
 - Overlapping subprograms.
- Let us know your thoughts using the QR code on Canvas

Quiz

1. Pen and paper; no compute or cellphone allowed.
 2. Turn in your answer sheet when you leave the classroom.
- **Q1 (Multiple Choice)** What's the correct correspondence of POS tags and the full name?
 - a) Verb -> VB
 - b) Noun -> DT
 - c) Adjectives -> NN
 - d) Adverb -> RB
 - **A1: a, d**
 - **Q2 (True/False)** The transition matrix is a symmetric matrix, namely, $A_{ij} = A_{ji}$ for any i, j
 - **A2: F**
 - **Q3 (Short answer)** What's the space complexity of storing the emission matrix, when there are $|S|$ unique tags and $|V|$ unique words?
 - **A3: $|S| \times |V|$**